

Lab 4: Sequence Diagrams

Project: eRegistrar

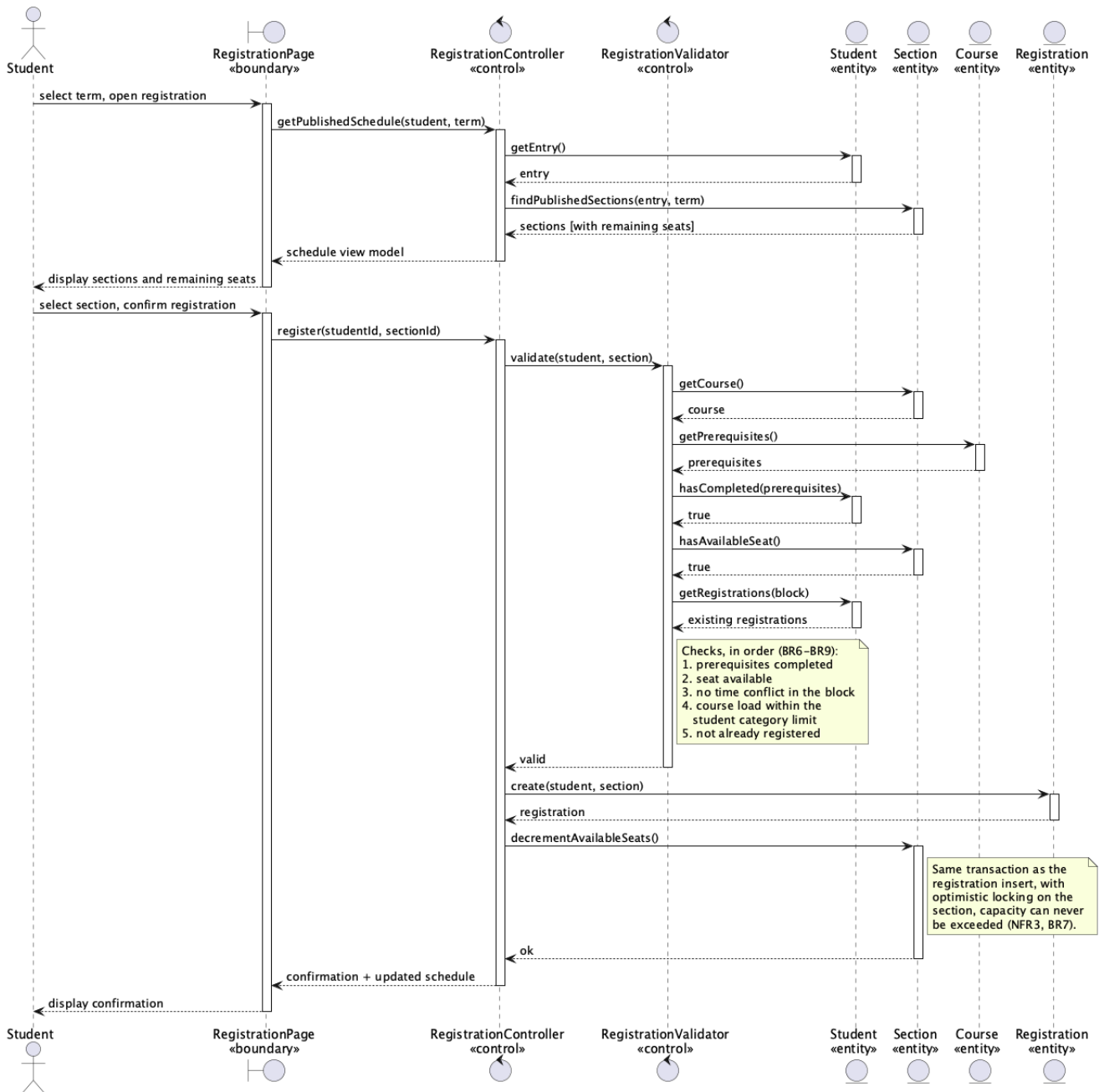
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Sequence diagrams for the two major use cases of the SRS (Lab 2), drawn at the analysis level. Every lifeline is stereotyped as «**boundary**», «**control**» or «**entity**», and messages flow actor → boundary → control → entity, never back up.

Stereotype	Role
« boundary »	Mediates between an actor and the system; holds no business rules.
« control »	Coordinates the use case and applies the business rules.
« entity »	Long-lived stored information.

UC4: Register for Course

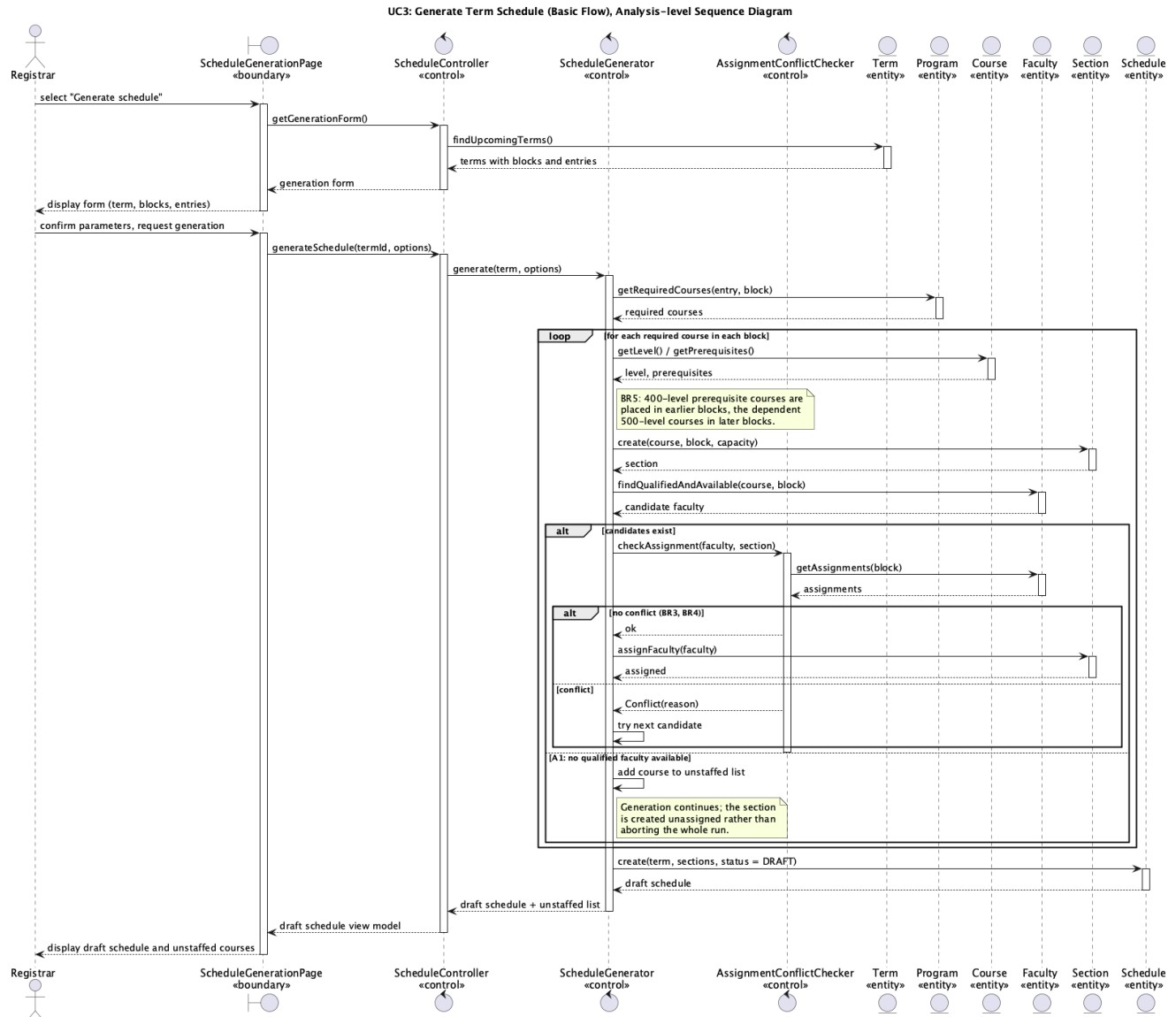
UC4: Register for Course (Basic Flow), Analysis-level Sequence Diagram



Class	Stereotype	Responsibility
RegistrationPage	boundary	Shows the schedule with remaining seats; collects the student's selection.
RegistrationController	control	Coordinates the use case; owns the transaction boundary.
RegistrationValidator	control	Applies BR6 to BR9: prerequisites, capacity, time conflict, course load.
Student, Section, Course, Registration	entity	The registration data.

The key detail is that the seat check and the seat decrement happen in the same transaction on **Section**, under optimistic locking. That's what makes NFR3 (capacity holds under concurrent registration) achievable.

UC3: Generate Term Schedule



Class	Stereotype	Responsibility
ScheduleGenerationPage	boundary	Generation form; displays the draft schedule and unstaffed courses.
ScheduleController	control	Coordinates the use case.
ScheduleGenerator	control	Determines required courses, creates sections, selects faculty (BR5).
AssignmentConflictChecker	control	Enforces BR3 (no double booking) and BR4 (only teachable courses).
Term, Program, Course, Faculty, Section, Schedule	entity	The scheduling data.

Alternate flow A1 shows up in the diagram: when no qualified, available faculty member exists, the section is created unassigned and the course is added to the unstaffed list instead of aborting the run.

Re-rendering

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java -jar tools/plantuml.jar -tpng Lab4_SequenceDiagrams/diagrams/*.puml
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